

Bahesty Monfared

I build things I don't yet know
how to build — then [write](#)
[down](#) how they work.

Final-year BSc student in Electrical and Electronic Engineering at AIUB.
Most of my work sits where hardware meets software: an ESP32 driving a
real pump, a solar array that has to hand over to the grid cleanly, a dataset
that has to be cleaned before it can tell you anything. Alongside that, I have
spent several years teaching — which is where I learned that a design is
only finished when somebody who wasn't in the room can follow it.

DEGREE	INSTITUTION	EXPECTED GRADUATION	CGPA
BSc, Electrical & Electronic Engineering	American International University–Bangladesh	December 2026	3.38 / 4.00

bahestymonfared@gmail.com

LinkedIn

+880 1834-211666

HOW I TAKE ON SOMETHING I'VE NEVER DONE BEFORE

01 understand what is being asked → 02 break it into requirements → 03 check what evidence I have →
04 name the gaps → 05 research what I don't know → 06 decide → 07 write down why

Selected work

Six projects · 2024–2026

ENGINEERING & HARDWARE

FINAL-YEAR CAPSTONE

AIUB, Faculty of
Engineering
Role: sensing, control &
firmware
Status: built and defended

HydroSense — IoT smart irrigation system

An automated irrigation controller for small farms. Soil moisture and flow sensors feed an ESP32, which drives the pump zone by zone instead of running on a fixed timer. It reports over GSM so it works on land with no Wi-Fi, and the interface is a bilingual progressive web app — a farmer can use it on a low-end phone with a weak connection.

- Designed the sensing and control side — soil moisture and flow sensor integration, relay-driven pump control, and the zone logic that decides when to water.
- Wrote the ESP32 firmware and dealt with the electrical problems that only appear once a real pump is attached, including separating the pump supply from the control electronics so the board stopped resetting on pump start-up.
- Taught myself progressive web app development from scratch — offline caching, an installable interface, and a bilingual layout for users who don't read English. None of that is in an EEE syllabus.
- Wrote the capstone defence book documenting the design decisions, the test results, and the reasoning behind each one.

ESP32Soil moisture & flow sensorsGSM moduleRelay controlC / C++

HTML / CSS / JS (PWA)

TEAM PROJECT · GROUP 2026.2.7

AIUB · four members
Role: embedded & data
pipeline
Prototype tested 19–27 Jul
2026

Hybrid solar–grid fast charging system with AI diagnostics

A charging system that draws from solar first and falls back to the grid, with MPPT tracking, a smart battery management layer for protection, and a machine learning pipeline that flags faults from live sensor data. An ESP32 acts as the edge node; a Raspberry Pi Zero 2 W runs the server and the model. Built as a working prototype, then written up as a six-chapter project book.

- **Built the embedded side** — ESP32 firmware split into sensing, control, display and network modules, reporting to the Raspberry Pi server.
- **Owned the diagnostics dataset** — 302,915 raw samples cleaned down to 301,696 usable ones, working through each cleaning decision rather than accepting the raw log as it came.
- **Compared simulated behaviour against measured hardware** and documented where the two disagreed and why, instead of reporting only the runs that matched.
- **Produced the documentation** — the six-chapter book, appendices of IC datasheets and source code, and a condensed IEEE-format summary paper drawn from it. Cleared plagiarism screening at 8% similarity with no integrity flags.

ESP32 Raspberry Pi Zero 2 W MPPT BMS Python ML pipeline C / C++

COURSE PROJECT · RENEWABLE ENERGY TECHNOLOGY

AIUB · group
Role: modelling & sizing
Output: IEEE-format report

Off-grid solar PV system design and simulation for Bangladesh

A full sizing and simulation study for a standalone solar installation under Bangladeshi irradiance and load conditions, ending in a 2S × 4P eight-module array configuration. The work covered load estimation, panel and battery sizing, and modelling how the system would behave before any of it was built.

- **Modelled the array in MATLAB and Simulink** and produced the I–V and P–V characteristics used to justify the chosen configuration.
- **Worked the sizing calculations by hand first** — irradiance, module output, series and parallel arrangement — so the simulation was checking my reasoning rather than replacing it.
- **Wrote it up in IEEE format**, which meant stating assumptions explicitly and citing a source for every figure I hadn’t measured myself.

MATLAB Simulink PV system sizing IEEE technical writing

RESEARCH & ANALYSIS

UNDERGRADUATE THESIS

AIUB · group research
Role: dataset evaluation &
method design
Status: in progress

Effect of colour and layout on user attention in web interfaces

A study of how visual design choices change where people actually look on a web page. Much of the work was evaluating existing eye-tracking and saliency datasets against our requirements, then designing our own experiments for the parts no dataset covered.

- **Evaluated published eye-tracking and saliency datasets** — UEyes, Imp1k, FiWI — comparing what each one measured against what the study needed. That is a requirements-matching exercise before it is a research one.
- **Helped design the experiments for the gaps:** A/B testing, five-second memory tests, blur tests, distraction tests and side-by-side ranking.
- **Worked in a field with no electrical engineering content at all**, which meant reading unfamiliar literature and learning another discipline’s vocabulary before I could contribute anything.

Research methodology Dataset evaluation Experimental design
Literature review

COURSEWORK · HUMAN-COMPUTER INTERACTION

AIUB

Role: sole author

Output: annotated presentation

Usability analysis of the Steam desktop application

A structured evaluation of a large, widely used desktop application against Don Norman's design principles — visibility, feedback, affordance, mapping, constraints and consistency — delivered as an annotated presentation.

- **Judged a messy real interface against a fixed checklist**, screen by screen, instead of offering a general opinion about whether the app was good.
- **Annotated the evidence directly on screenshots** so a reader could check the conclusion themselves rather than take my word for it.

Usability heuristics

Structured evaluation

Annotated presentation

TEACHING MATERIAL

HSC ICT curriculum

Role: built from scratch

Animated HTML & web design lesson set

A teaching resource for the HSC ICT web design unit, built as an animated presentation with the demonstrations coded rather than described — so students see the markup and the rendered result change together instead of reading about it.

HTML

CSS

PowerPoint animation

Curriculum design

Experience

Teaching · customer-facing work

ACADEMIC TEACHER

Carupath Academic Coaching
IT department

Oct 2022 – Sep 2025

Teaching technical subjects to students starting from zero

Three years teaching IT subjects to students with no background in them. Teaching a subject to somebody who has never seen it is the fastest way to find out where your own explanation breaks down — and that habit carries straight into writing a design document somebody else has to act on.

- **Built lesson material from the ground up** for students at mixed levels within the same class.
- **Explained technical judgements to non-technical listeners** daily — the same task as defending a design decision to a client or a panel.

VOLUNTEER TEACHER

Bidyanondo Foundation
Jan 2019 – Dec 2023

Five years teaching in a resource-limited setting

Volunteer teaching with limited materials and a wide spread of ability in one room, which meant rebuilding the same explanation several ways until it landed for everybody rather than for the fastest student.

BRAND PROMOTER

ACI Limited
Twinkle Baby Pant Diaper
Feb 2023 – Nov 2023

Direct customer-facing sales and product communication

Face-to-face promotion work for a consumer brand — a different communication discipline from a classroom or a technical report, and the one that taught me how quickly you lose someone who didn't ask to be talked to.

Education

BSC – IN PROGRESS

Expected December 2026

Electrical & Electronic Engineering

American International University–Bangladesh (AIUB), Dhaka · CGPA 3.38 / 4.00 (ongoing)

Power systems

Renewable energy technology

Numerical methods

Web technologies

Human-computer interaction

HSC – SCIENCE

2021 · GPA 5.00 / 5.00

Mazid Jarina Foundation School and College

Skills			Tools I have actually shipped something with
DESIGN & SIMULATION	PROGRAMMING	HARDWARE & EMBEDDED	
AutoCAD · NI Multisim (all versions) · PSpice · MATLAB · Simulink · Proteus	C · C++ · Python (ML pipeline work) · HTML / CSS / JavaScript	ESP32 · Raspberry Pi Zero 2 W · sensor integration · relay and pump control · GSM modules · MPPT and BMS concepts	
DOCUMENTATION	DOMAINS	LANGUAGES	
IEEE-format technical writing · structured project books · MS Word / Excel / PowerPoint	Power systems · renewable energy · embedded IoT · numerical methods · web technologies · human–computer interaction	Bangla (native) · English (professional working proficiency)	

Get in touch

Open to internships and graduate roles

I’m looking for internships and entry-level roles where I can keep working on real hardware and real documentation — embedded and IoT, power and renewable energy, or engineering roles that involve explaining technical work to people outside the team. I’m based in Dhaka and available for on-site, hybrid or remote work.

bahestymonfared@gmail.com

+880 1834-211666

LinkedIn

One thing stated plainly: I am in my final year and graduate in December 2026 — I don’t hold the degree yet.

I’d rather say that here than have it come up later. Everything on this page is work I have already done.